

SpeechCARE: A Novel Multimodal Fusion Transformer Model for Early Detection of Cognitive Impairment–NIA Alzheimer's Challenge

SpeechCARE Team

- **Maryam Zolnoori**^{a,b}, PhD. Assistant Professor - mz2825@cumc.columbia.edu
(Speech processing Lab)
- **Hossein Azadmaleki**^a, M.Sc. Computer Science - h.azadmaleki@gmail.com
(Development of the SpeechCARE pipeline, Report drafting, Project management)
- **Yasaman Haghbin**^a, M.Sc. Computer Science
(Transformer models, Bias analysis, Error analysis)
- **Mohammad Javad Momeni Nezhad**^a, M.Sc. Computer Science
(Large language models)
- **Sina Rashidi**^a, M.Sc. Computer Science
(Data augmentation techniques, Signal processing)
- **Ali Zolnour**^a, M.Sc. Computer Science
(Performance analysis, Results visualization)

^a Columbia University Irving Medical Center, New York, NY 10032, United States

^b School of Nursing, Columbia University, New York, NY 10032, United States

Address: 560 W 168th St, New York, NY 10032

Phone: 317-515-1950

E-mail: mz2825@cumc.columbia.edu

Other team members

Mehdi Naserian, Elyas Esmaeili, Mahdis Hosseini (Hand-crafted features modeling, Data preprocessing)

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A. ABSTRACT. Alzheimer’s disease and related dementias (ADRD) affect one in five adults over 60, presenting a significant public health concern.^{1–3} Despite nationwide efforts, over half of individuals with cognitive decline, including mild cognitive impairment (MCI) and ADRD, remain undiagnosed.^{4–6} Recent studies showed that speech-based assessments hold promise for early detection: phonetic motor planning deficits impair vocal tract control (e.g., vocal folds), altering acoustic features like pitch and tone.⁷ Memory and language impairments disrupt language organization, leading to syntactic and semantic errors and reduced fluency.^{7,8,9} The studies mostly focused on conventional speech-processing pipelines with hand-crafted features (e.g., eGeMAPS¹⁰) or general-purpose audio classification models (e.g., YAMNet¹¹, VGGish¹²). However, these approaches often exhibit suboptimal performance and limited generalizability across diverse language settings. To address these gaps, we introduce SpeechCARE, a multimodal speech processing pipeline leveraging underutilized pretrained, multilingual acoustic and linguistic transformer¹³ models to capture nuanced acoustic and linguistic cues associated with cognitive impairment. Inspired by Mixture of Experts (MoE) paradigm,¹⁴ its core architecture is a novel multimodal fusion architecture that dynamically weights transformer-driven acoustic and linguistic features for effective integration, enhancing performance and generalizability across different speech production tasks (e.g. story recall, sentence reading). This fusion mechanism has the capability of seamless integration of additional data (e.g., social determinants,¹⁵ MRI¹⁶), boosting its sensitivity across the cognitive impairment spectrum. By leveraging pretrained transformer models, SpeechCARE can tackle challenges posed by small sample sizes, thereby enabling the inclusion of often-overlooked linguistic diversity. SpeechCARE’s robust preprocessing pipeline includes automatic transcription, LLM (Large Language Model¹⁷)-assisted data anomaly detection, and LLM-assisted speech task identification. SpeechCARE’s explainability framework visualizes each modality’s contribution to decision-making, highlighting linguistic and acoustic cues linked to cognitive impairment through a novel SHAP-based approach and LLM-based reasoning. SpeechCARE achieved AUC = 0.88 and F1 = 0.72 in distinguishing cognitively healthy, MCI, and AD. Specifically, for MCI detection (versus control), AUC = 0.90 and F1 = 0.62. Bias analyses using equal opportunity and average odds showed no significant differences across demographics except for age over 80. Various techniques (e.g., oversampling, weighted loss) were used for bias mitigation. Future Directions. SpeechCARE is accessible, non-invasive, and cost-effective for real-world care settings. Our immediate goal is to validate its performance on patient-clinician communications from VNS Health and Columbia University’s Alzheimer’s Disease Research Center, focusing on underrepresented groups (Hispanic, Black) in New York City (NYC). With support from Columbia’s visualization center, we aim to evaluate its explainability in clinician-centered design for Electronic Health Record (HER) integration, enabling timely diagnosis of cognitive impairment and early intervention for diverse populations.

B. METHODOLOGY

B.1. Dataset characteristics. The PREPARE Challenge comprises 2,058 participants (1,646 in the training set and 412 in the test set), including 1,140 healthy controls, 268 individuals with mild cognitive impairment (MCI), and 650 individuals with Alzheimer’s disease (AD) (Figure 1). Participants spoke three languages: English (1,655), Spanish (360), and Mandarin (43). Gender was reported as female or male, with females comprising 1,219. Ages ranged from 46 to 99 years (avg: 75.13 ± 8.65). Education was reported both numerically and categorically with 35.23% missing data. Race was reported for less than 8% of the participants (92.61% missing).

B.2. Preprocessing components. (a) **Demographic information preparation:** To facilitate model bias analysis and effectively incorporate demographic data, we grouped age into three categories following common study conventions: mid-life adults (46–65, 12%), older adults (66–80, 52.7%), and elderly (80+, 35.3%). To normalize education without missing values, we categorized it into four levels: no/elementary (11.61%), high school (24.98%), technical/undergraduate (16.72%), and advanced/graduate (11.47%). We addressed missing

values (35.23%) using Iterative Imputer,¹⁸ a multivariate regression method modeling them from age, gender, and language as input features.

(b) Data quality evaluation and improvement: High-quality audio data is essential for encoding subtle acoustic cues particularly for MCI detection.¹⁹ However, the recordings included different types of stationary (e.g., recording device’s hum) and dynamic noises (e.g., nearby conversations). To measure noisiness, we used different techniques including Signal-to-Noise Ratio and Spectral Flatness Measure, but due to inconsistent preprocessing by corpora developers, the outcomes were not strongly correlated with human perception. Therefore, we used a qualitative method, Mean Opinion Score,²⁰ where two experts (HA, MZ) listened to 5% of the data (stratified by demographic and diagnostic classes) and rated quality on a 1–5 Likert scale.²¹ Findings showed 62.4% severely noisy (scores 1–2), 14.7% moderately noisy (score 3), and 22.9% slightly noisy (scores 4–5). To safely mitigate the noise, we then applied a low-pass filter²² at 8,000 Hz to remove high-frequency noise while preserving human speech frequency components.

(c) Automatic transcription: To incorporate linguistic cues into the SpeechCARE pipeline, we utilized the latest version of Whisper-Large,²³ an open-source, multilingual automatic speech recognition system that achieved state-of-the-art (SOTA) performance on standard benchmarks. Whisper is able to detect the language of input speech. To evaluate Whisper’s transcription accuracy, we used a stratified random sample (by demographic and diagnostic classes) comprising 5% of English, 10% of Spanish, and 15% of Mandarin audio data, ensuring sufficient sample sizes for each language. Transcriptions were compared to human-generated transcripts to calculate the word error rate²⁴ (WER): English = 0.12, Spanish = 0.27, and Mandarin = 0.32.

(d) Speaker diarization: SpeechCARE’s diarization module builds on our previous study, which used a machine-learning algorithm to automatically separate the patient and clinician in recorded conversations. Based on our prior studies and initial findings of this study, incorporating available clinician data from certain corpora improved overall performance. Therefore, clinician speech was included in model training whenever available in the audio files of this dataset.

(e) LLM-Assisted data anomaly detection: During the evaluation of the Whisper transcriptions, we discovered that some audio files contained only the clinicians’ speech, likely due to errors in speaker diarization performed by the corpus developers. To systematically identify these cases, we employed LLaMA 405B²⁵—an open-source LLM comparable to GPT-4 performance—with prompt engineering²⁶ and few-shot learning²⁷, guided by iterative analysis of the model’s output. This approach revealed 22 audio files consisting solely of clinicians’ speech. We removed these files from the training process to prevent potential confusion in the model.

(f) LLM-assisted speech task identification: Speech production task type was not available in the dataset, yet identifying it is important for model development and bias/error analysis. Each task requires distinct cognitive effort and yields different acoustic and linguistic cues for cognitive impairment detection. We used LLaMA 405B with prompt engineering and few-shot learning, guided by iterative expert analysis (HA, MZ), to refine the LLM prompt for task identification. The same experts then verified 10% of the assigned tasks (stratified by demographics and diagnostic classes), confirming 98% accuracy. Errors arose mainly from truncated or insufficient transcripts. See **Figure 1** for statistics on the identified tasks.

(g) Task homogeneity and bias introduction: According to these analyses, we observed that (1) all Spanish speakers performed only the sentence reading task, which may not provide

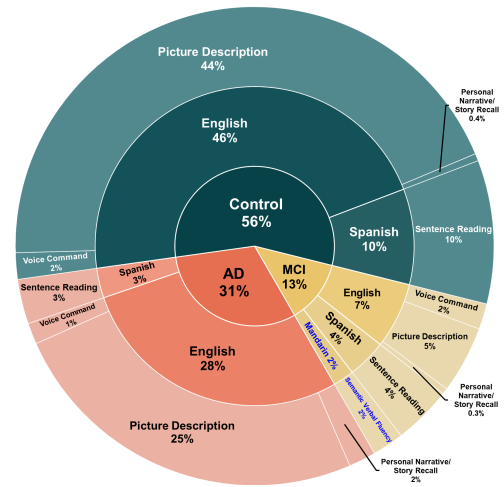


Figure 1. Statistics on diagnosis classes, language, and speech tasks.

sufficient temporal cues linked to cognitive impairment compared to picture description and story recall. This limitation introduces bias and may reduce pipeline performance for these participants; (2) all Mandarin speakers had MCI in both the training and test sets, inflating speech pipeline performance because the system simply learns to classify all Mandarin speakers as having MCI.

B.3. Building speech processing models. Our team has extensive experience developing speech processing pipelines for cognitive impairment detection, highlighted in news media^{28–31} and high-impact journals.^{32–41} Overall, no single approach consistently yields optimal performance due to variations in speech corpus characteristics (e.g., speech task, language) and audio data quality. Consequently, we investigated various linguistic and acoustic processing methods for this dataset to build SpeechCARE pipeline on the top-performing and most generalizable models (See **Table 1** for brief information and results).

(a) Hand-crafted speech processing models: We developed three pipelines based on hand-crafted acoustic and linguistic features (see more information and results in **Table 1**): (1) SpeechDETECT:³² Drawn from a systematic review of studies focused on speech and language for MCI detection, we built SpeechDETECT, providing a comprehensive, explainable set of acoustic and temporal features of speech; (2) eGeMAPS-Based model: We built this model on the eGeMAPS V02⁴² parameter set (provided by the challenge organizer), using advanced time-series analysis;¹³ (3) LinguisticDetect:⁴⁰ Drawn from previous speech-related studies for MCI detection, we built this model by quantifying lexical richness, syntactic complexity, semantic coherence, and psychological cues linked to cognitive impairment using natural language processing methods.

(b) Speech (acoustic) and linguistic pre-trained transformer models: Pre-trained speech transformer models (e.g., HuBERT,⁴³ Wav2Vec 2.0⁴⁴) are powerful in capturing long-term acoustic and temporal dependencies. However, they have been underutilized in cognitive impairment related studies due to the lack of effective fine-tuning procedure on small speech datasets. For this dataset, we evaluated (1) SOTA pre-trained speech-transformer models—Wav2Vec2 XLS-R,⁴⁵ mHuBERT;⁴⁶ (2) pre-trained linguistic-transformer models—mBERT⁴⁷ and XLM-RoBERTa⁴⁸ (commonly used models) as well as mGTE⁴⁹ and BGE⁵⁰ (SOTA models). We selected these models due to multilingual pre-training and high generalizability across diverse languages. Our initial analysis on this dataset showed that multilingual models (e.g., mHuBERT, mBERT) generally outperformed monolingual counterparts (HuBERT, BERT⁵¹), even for English-speaking participants; (3) Audio Spectrogram Transformer⁵² (AST). SOTA Speech transformer models primarily were designed for automatic speech recognition. Therefore, we also tested the AST model, a common model for audio classification trained on YouTube audio. All these transformer models were fine-tuned using procedure of fine-tuning detailed in section B.5.

(c) LLM-based models: We investigated LLMs, which extend transformer architectures by greatly increasing model parameters and training data size. Prior work showed that LLMs excel at recognizing linguistic cues linked to cognitive impairment with minimal fine-tuning.^{53,54} For local, HIPAA-compliant deployment, we fine-tuned: LLaMA 3²⁵ (versions 3.1 70B, 3.1 8B, 3.2 3B) and Ministral 8B⁵⁵ using QLoRA⁵⁶ (a parameter-efficient, low-rank adaptation method) combined with a linear classifier on the final-layer embeddings to generate probabilities.

B.4. Evaluating speech processing models. We evaluate the performance of the processing pipelines by (a) creating a validation dataset by randomly selecting 20% of the training dataset, stratified using demographic and diagnostic classes—a standard practice in deep learning algorithms development; (b) standard metrics for multi-class evaluation: AUC-ROC (Area Under the Curve - Receiver Operating Characteristic, one-vs-rest approach for three classes), F1-score (harmonic mean of precision and recall) using the micro-average, which accounts for class imbalance, and multi-class log loss as instructed by challenge organizers. See **Table 1** for performance of the speech processing models on the validation set.

Table 1. Summary of acoustic and linguistic models and their performance

Models Characteristic				Validation Result		
1.1. Acoustic Hand-Crafted Features						
Models		Description		Loss	AUC	F1
SpeechDetect	6848 features encoding Frequency, Spectral, Voice Quality, Intensity, Signal Complexity, Rhythmic Structure, Fluency, and Speech Dynamics aspects of speech			0.801	0.80	0.62
eGeMAPS	88 features encoding Frequency, Spectral, Voice Quality, and Intensity aspects of speech			0.879	0.77	0.57
1.2. Linguistic Hand-Crafted Features						
Model		Description		Loss	AUC	F1
LinguisticDETECT	N features encoding lexical density, syntactic structure, semantic and psycholinguistic cues			0.900	0.70	0.63
1.3. Speech Transformer Models						
Transformer	Size	Training source	input	Loss	AUC	F1
Wav2vec2 XLS-R	300 M	436K hours /128 languages	Waveform	0.849	0.78	0.56
mHuBERT	96 M	90K hours /147 languages	Waveform	0.738	0.84	0.66
AST	87 M	2M 10-second audio clips	Spectrogram	0.803	0.80	0.60
1.4. Linguistic Transformer Models						
Transformer	Size	Training source		Loss	AUC	F1
XLM-RoBERTa	600 M	~295 billion tokens across 100 languages		0.841	0.80	0.64
BGE	567 M	~1.2 billion text pairs across 194 languages		0.806	0.81	0.66
mGTE	305 M	~1028 billion tokens across 75 languages		0.789	0.83	0.67
mBERT	177 M	~2.5 billion tokens across 104 languages		0.803	0.82	0.66
1.5. Large Language Models						
LLM	Size	Availability	Fine-Tuning Method	Loss	AUC	F1
LLaMA 3.1	70 B	Open weight	QLoRA	0.787	0.82	0.68
LLaMA 3.2	3 B	Open weight	QLoRA	0.855	0.80	0.65
LLaMA 3.1	8 B	Open weight	QLoRA	0.811	0.81	0.66
Ministral	8 B	Open weight	QLoRA	0.817	0.81	0.64

B.5. Fine-tuning transformer models. The fine-tuning procedure includes two components:

(a) Fine-tuning speech transformers: Fine-tuning speech transformers on raw waveforms is resource-intensive, mainly due to the quadratic complexity of multi-head attention as input length grows. This restricts the maximum audio duration the model can handle. Our previous work showed that models like HuBERT and Wav2Vec 2.0 perform best with audio segments under 15 seconds (s). To handle longer audio, we split each file into 5-second segments with 25% overlap, creating up to seven segments for 30 s files. Shorter files (<30 s) were padded. Next, the speech transformer encodes each segment into a sequence of 250 embedding vectors (each representing 25 milliseconds of audio with 20% overlap). Concatenating embeddings from all segments yields up to 1750 vectors (7×250). To capture long-range dependencies across segments, we prepend a randomly initialized speech [CLS] embedding (1751 vectors total) and input the sequence into a Customized Lightweight Multi-Head Self-Attention (CLMS) encoder

(**Figure 2** for more detailed structure). The CLMS encoder, with two blocks and four heads each, has fewer parameters, making it well-suited for our small dataset. Finally, the processed [CLS] embedding passes through a feedforward layer to produce outcome scores for diagnostic classes. This training procedure helps the speech [CLS] embedding capture robust acoustic and temporal features of the entire audio.

(b) Fine-tuning procedure of linguistic transformers: We adopt the standard approach of fine-tuning linguistic transformers on transcriptions, passing their [CLS] embeddings through a feedforward network to generate screening scores.

B.6. Multimodal fusion models for speech processing.

(a) Fusion methods: We investigated the performance of four fusion models to combine different modalities including speech (acoustic), linguistic, and demographic features: (1) Intermediate Fusion⁵⁷ (concatenating modality embeddings), (2) Late Fusion⁵⁸ (combining class predictions per modality), (3) Cross-Attention Fusion⁵⁹ (capturing inter-modality dependencies through attention mechanisms), and (4) Adaptive Gating Fusion (adaptive fusion of modalities using learnable attention gates).

Table 2. Result of multi-modal fusion models

Fusion Method	Validation Result			Test Result		
	Loss	AUC	F1	Loss	AUC	F1
Intermediate Fusion	0.722	0.85	0.68	0.720	0.86	0.68
Late Fusion	0.698	0.82	0.71	0.670	0.87	0.71
Cross-Attention	0.747	0.85	0.68	0.738	0.86	0.69
<i>Submitted Model to Model Arena Phase</i>						
Adaptive Gating	0.754	0.84	0.67	0.655	0.88	0.72

(b) Building multimodal models: Based on the outcome of speech processing models (section B.3), mHuBERT and mGTE provided the highest performance on the validation dataset (**Table 1**) and were used for building the multimodal models using the four approaches (results in **Table 2**). Since Adaptive Gating Fusion (AGF) provided the highest performance, we selected it for building our final model and provided a brief description in section B.6.c.

(c) Adaptive Gating Fusion (Figure 1): Inspired by Late Fusion and Mixture-of-Experts (MoE) paradigm¹⁴, AGF uses a gating network to dynamically determine the contribution (weight) of each modality. First, the [CLS] embeddings from each modality are passed through a feed-forward layer (Tanh activation) to produce hidden representations. These representations are then passed to separate feed-forward layers to compute individual outcome scores for each modality. Additionally, concatenation of modalities' representations is passed to the gating network (a single feed-forward layer following a Softmax layer) to generate modality weights (**Figure 2**). The final outcome prediction is a weighted sum of modality-specific outcome scores, guided by the gating network's weights.

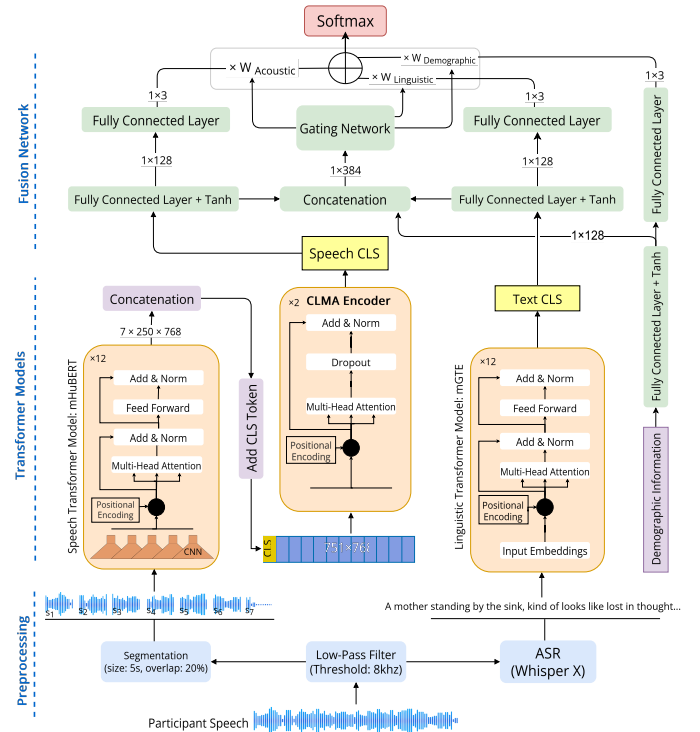


Figure 2. SpeechCARE's core model architecture.

C. PERFORMANCE ANALYSIS

(a) Performance metrics. We used AUC-ROC (one-vs-rest approach) to evaluate how well the model distinguishes each class. We computed micro-average and weighted-average AUC: micro-average aggregates decisions across all classes equally, while weighted-average accounts for class imbalance. We also calculated Precision-Recall (PR) curves using the same approach. Additionally, we introduced Information Gain curves to measure how much uncertainty the model's predictions reduce compared to random guessing (Figures 3.A–3.C).

(b) Performance of SpeechCARE's models (Figures 3.A–3.C). **(a) SpeechCARE- AGF** with adaptive gating fusion (AGF) on mHuBERT, mGTE, and demographics achieved AUC=87.59 (Micro) and PR=77.17 (Micro). The cumulative gain curve showed that by targeting the top 40% of the population, ~80% of True diagnoses were captured;

(b) SpeechCARE-Whisper replaces mHuBERT with Whisper-Medium (using encoder embeddings). Overall, Whisper dominated mGTE and demographics, causing the model to assign minimal/zero weight to them. This likely happened because Whisper was trained on a larger labeled dataset (680k h) compared to mHuBERT's smaller unlabeled set (90k h). Nonetheless, SpeechCARE-Whisper's performance did not significantly differ from SpeechCARE-AGF, indicating the robust AGF architecture for speech processing.

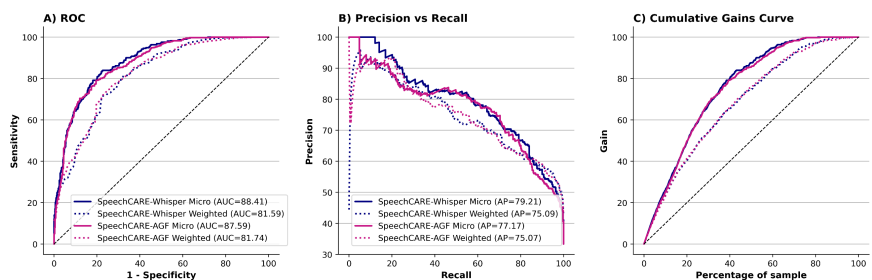


Figure 3. SpeechCARE -AGF & Whisper

(c) Performance of SpeechCARE-AGF and SpeechCARE-Whisper for MCI detection (Figures 4.A–4.C). Similarly to C.2, there was no significant difference between both models for MCI detection (MCI class vs. Control class). Overall, the result is promising with AUC-ROC = 90.50, RP = 68. **After threshold optimization**, we achieved F1= 0.62, recall=0.73, precision=0.54 for MCI, indicating a good performance of SpeechCARE for early detection.

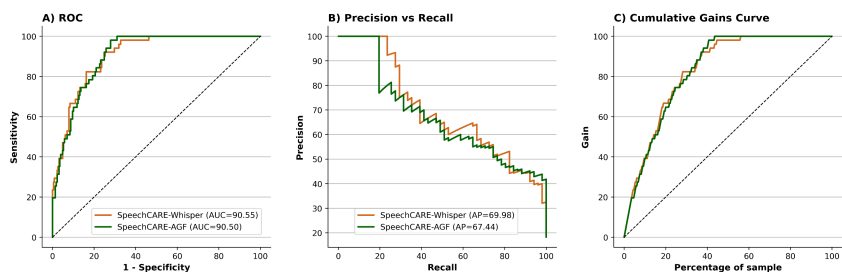


Figure 4. SpeechCARE -AGF & Whisper: MCI vs. Control

(d) Error analysis (Figure 5). We observed no false negatives (misclassified MCI/AD as controls) in Mandarin speakers because all Mandarin speakers in the dataset had MCI. Spanish speakers showed the highest false negatives, likely reflecting the limited ability of sentence-reading tasks to capture acoustic/linguistic cues (see B.2.(f)). English speakers primarily completed picture-description and narrative tasks. For these participants, we used a T-test ($p=0.05$) to compare acoustic (fundamental frequency, jitter, shimmer, intensity, MFCC) and linguistic (lexical richness, repetition, fillers, sentence structure) features between false negative class and control class (true negative). No significant differences were found in either feature set between these two groups. This may indicate that language-related brain regions may be less affected in the false-negative group. Incorporating additional modalities (e.g. medical history, MRI) may improve SpeechCARE sensitivity.

True Labels	Control	190	30	9
	MCI	10	41	0
	AD	47	19	66
		Control	MCI	AD

Figure 5. SpeechCARE-AGF Confusion Matrix. Note that the Thresholds were adjusted for three classes.

D. BIAS ANALYSIS

(a) Metrics. We used Equality of Opportunity⁶⁰ (EOO) and Average Odds, two of the most common bias metrics. For demographic data (**Figure 6.A**), no significant bias was observed except for the age-over-80 group. For languages (**Figure 6.A**), bias stemmed from dataset constraints: (1) all Mandarin speakers had only MCI, inflating EOO; (2) all Spanish speakers only completed sentence-reading tasks, limiting critical speech cues for detection.

(b) Methods. We applied oversampling as a preprocessing method, incorporating Frequency Masking⁶¹ and Voice Conversion⁶² for speech augmentation, alongside in-processing methods such as Adversarial Debiasing⁶³ and reweighting⁶⁴. For Spanish speakers, SpanBERTa (a Spanish transformer) was replaced with mGTE.

(c) Results of mitigation. Among oversampling techniques, frequency masking outperformed voice conversion. Focal loss slightly reduced biases, while other in-processing methods showed no improvement. Replacing SpanBERTa with mGTE significantly mitigated bias. Overall, these methods improved EOO by 5.17% for the age-over-80 group and by 16.36% for Spanish speakers (**Figure 6.B**), demonstrating the effectiveness of our bias mitigation approach.

E. FUTURE DIRECTIONS

(a) Multimodal integration of speech, biomarkers, EHR data, social determinants from Columbia Alzheimer’s Disease Research Center (ADRC). Speech-only markers may underperform when language-related brain regions are less affected. We are collaborating with Columbia ADRC to collect speech data from racially diverse groups in NYC. We plan to (1) fine-tune SpeechCARE on the multilingual dialects, (2) apply the SpeechCARE’s adaptive gating fusion to measure speech’s added value for early detection, (3) co-train speech embeddings with imaging biomarkers to uncover cross-modal links between speech features and brain changes.

(b) Fine-tuning SpeechCARE on routine patient-clinician communication and enhancing explainability framework. Speech production tasks overlook vital communication cues. We are collaborating with VNS Health (the largest home healthcare agency in the U.S.) to collect routine patient-clinician communications from a racially diverse population. We plan to (1) Fine-tune SpeechCARE on the communications data and (2) refine SpeechCARE’s Explainability Framework in a clinician-centered design, preparing it for efficacy trial and EHR integration.

(c) Longitudinal monitoring of speech. A single speech recording overlooks gradual cognitive decline. In partnership with Columbia Data Visualization Center, we are developing “SpeechCARE Lite,” a mobile app for recording speech samples over time. We plan to (1) integrate time-series models (e.g., temporal transformers) for longitudinal analysis; (2) improve SpeechCARE Explainability to provide insights on cognitive changes over time.

(d) Improving pre-and in-processing components of SpeechCARE. Speaker diarization errors, transcription bias, and noise in audio data remain problematic. We plan to (1) use advanced deep learning methods for noise reduction, (2) fine-tune Whisper to reduce bias in racially diverse populations, (3) develop LLM-based speaker diarization, and (4) extend SpeechCARE’s fusion model as larger datasets become available.

Figure 6.A. Equality of Opportunity before bias mitigation

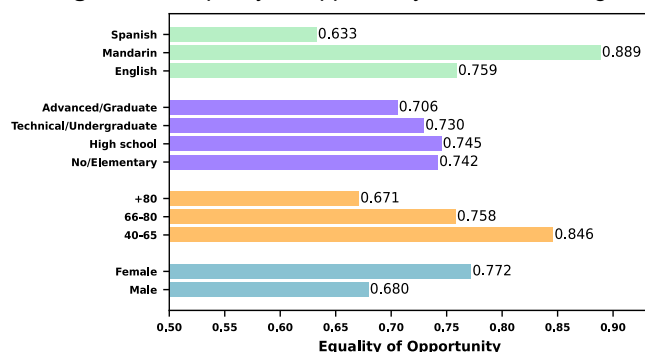
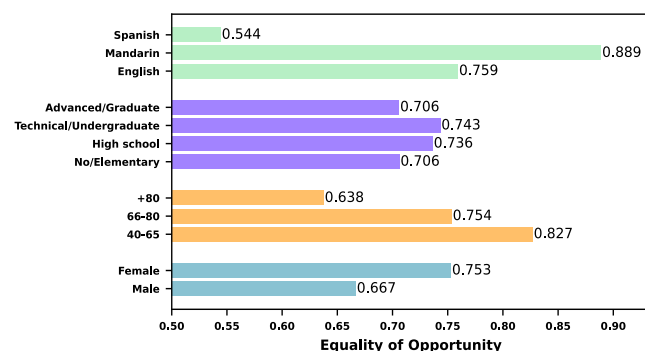


Figure 6.B. Equality of Opportunity after bias mitigation



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